Do LLMs Know When They're Making Things Up?

Hallucination detection on HaluEval QA

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Outline

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Problem

- A hallucination is a fluent, confident answer that is factually incorrect.
- The text itself gives no signal, so detection must rely on other signals.
- Three detectors are evaluated:
  - **Task 1:** the model's token probabilities (log-probability, entropy)
  - **Task 2:** SelfCheckGPT, i.e. consistency with 20 sampled answers (BERTScore, NLI)
  - **Task 3:** a logistic regression combining the Task 1 and Task 2 scores
- Metric: AUC-PR, with answer length as a reference baseline throughout.

Data and model

- **Data:** 200 of 10,000 HaluEval QA pairs (seed 42).
- Each pair gives a correct answer (label 0) and a ChatGPT-written hallucinated answer (label 1): 400 answers.
- **Model:** Qwen2.5-3B-Instruct (float16) on one RTX 5060 Ti (16 GB).
- Closed-book prompt: the `knowledge` paragraph is not given.

AUC-PR

$$\text{AUC-PR} = \sum_k (R_k - R_{k-1})\, P_k$$

Random detector: 0.5 (half of the answers are hallucinated).

Length difference

Correct answers: median 3 tokens. Hallucinated answers: median 14.5 tokens. The hallucinated answer is longer in 194 of 200 pairs; length alone gives AUC-PR **0.972**.

Deviations from the specification

- **Model size:** a 3B model, so that all steps fit on one 16 GB GPU.
- **Sample size:** 200 of 10,000 questions, since Task 2 needs 20 samples per question (4,000 generations).
- **NLI model:** `microsoft/deberta-v3-large-mnli` does not exist on Hugging Face; `potsawee/deberta-v3-large-mnli`, loaded by the official code, is used.
- **Library compatibility:** `selfcheckgpt` 0.1.7 calls a function removed in transformers 5; a patch restores it (README examples reproduced to six decimals).
- **Missing BERTScore:** four answers have no score (all samples were single words); imputed with the training median in Task 3.
- **Split by question:** both answers of a question stay in the same partition.

Task 1: Token probabilities

- Forced decoding: prompt x and answer $a = (a_1, \ldots, a_T)$ in one forward pass.
- Mean log-probability and mean entropy over the answer tokens:

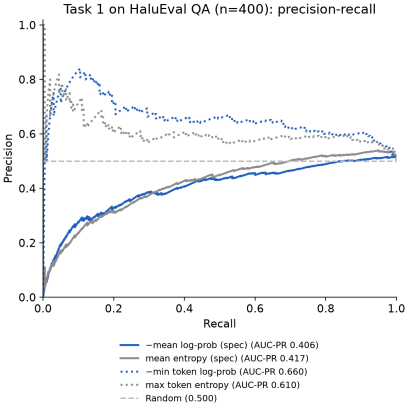
$$\bar{\ell}(a) = \frac{1}{T} \sum_{t=1}^{T} \log p(a_t \mid x, a_{<t}), \qquad \bar{H}(a) = \frac{1}{T} \sum_{t=1}^{T} \Big(- \sum_{v} p(v \mid x, a_{<t}) \log p(v \mid x, a_{<t}) \Big)$$

- Hallucination scores: $-\bar{\ell}$ and $\bar{H}$.
- Also reported: minimum token log-probability, maximum token entropy.

Task 1: Results

Score (all 400 answers)	AUC-PR	AUROC
$-\bar{\ell}$ (specified)	0.406	0.381
$\bar{H}$ (specified)	0.417	0.419
$-$minimum log-prob	0.660	0.697
maximum entropy	0.610	0.643
Random	0.500	0.500
Answer length only	0.972	0.971

- Both specified scores are **below chance**.
- The hallucinated answer has the higher mean log-probability in 63% of pairs.



Task 1: Why the mean log-probability fails

Answer	Token	$\log p$	Top prediction
Correct	`no`	-16.30	`No`
Hallucinated	`Yes`	-15.00	`No`
	`, both documentaries`	≈ 0	same
	`were`	-6.84	`Tr`
	`directed by`	0.00	same
	`Johannes`	-15.64	`Liz`
	`Gren`	-10.52	`Roberts`
	`zfurthner.`	≈ 0	same
Mean: correct / hallucinated		-16.30 / -3.48	

Pair 1290: *"Do the documentaries Traceroute and War/Dance have the same director?"*

- The model knows the answer: all 20 samples begin with "No".
- The lowercase token "no" is improbable (-16.3).
- Nine easy tokens raise the hallucination's mean to -3.48.
- The mean reflects surface form, not correctness.

Task 2: SelfCheckGPT

- $N = 20$ samples per question from the same model (temperature 1.0, top-$p = 1$, at most 48 new tokens, seed $42 +$ question id).
- Scored with the official `selfcheckgpt` package (0.1.7):

$$S_{\text{BERT}}(r) = 1 - \frac{1}{N} \sum_{n=1}^{N} \max_{k} \text{BERTScore-F1}\big(r,\, s_k^n\big)$$

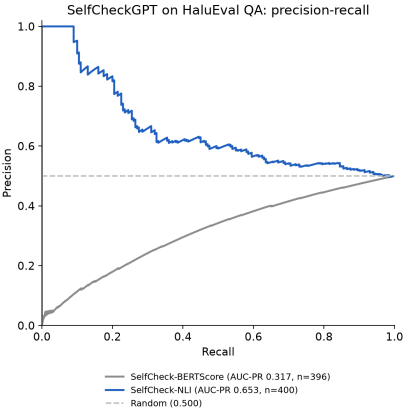
$$S_{\text{NLI}}(r) = \frac{1}{N} \sum_{n=1}^{N} P\big(\text{contradiction} \mid r,\, S^n\big)$$

- s_k^n: k-th sentence of sample S^n.
- BERTScore: `roberta-large` (layer 17) with baseline rescaling.
- NLI: DeBERTa-v3-large fine-tuned on MNLI. No model is trained in this task.

Task 2: Results

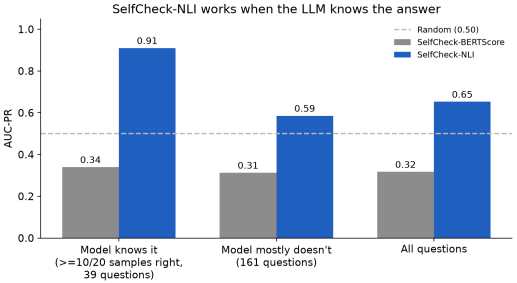
SelfCheck	AUC-PR	AUROC	Pairs
BERTScore ($n = 396$)	0.317	0.075	6%
NLI ($n = 400$)	0.653	0.614	67%
Random	0.500	0.500	50%

- "Pairs": hallucinated answer scored higher.
- NLI is above chance; precision 0.83 at 20% recall.
- BERTScore is **inverted**: the correct answer looks more suspicious in 94% of pairs. Sign-flipped AUC-PR: 0.918.
- Cause: word overlap with the samples depends on answer length.



Task 2: Effect of model knowledge

- SelfCheckGPT assumes that the model knows the answer.
- Only 42% of questions have at least one sample containing the correct answer.
- Only 39 questions (20%) have at least 10 of 20.
- On these 39 questions NLI reaches **0.91**; on the other 161 it reaches 0.59.
- Stable: 0.89 and 0.93 with thresholds of 5 and 15 samples.



Task 2: Examples

Pair 7787 (correct detection by NLI)

Q: "Woh Ladki Hai Kahan" ... released in which year? 18 of 20 samples say *2001*.

Correct: *2001* $S_{\text{NLI}} = 0.21$, $S_{\text{BERT}} = 1.06$

Hallucinated: *"Woh Ladki Hai Kahan" was released in 2000.* $S_{\text{NLI}} = 0.98$, $S_{\text{BERT}} = 0.42$

Pair 117 (BERTScore inverted)

Correct: *Roger Christian* $S_{\text{BERT}} = 1.10$

Hallucinated: *Jake David Shapiro was best known for being the screenwriter of a film directed by Steven Spielberg.* $S_{\text{BERT}} = 0.34$

The samples restate the question, so the long answer shares most of their words.

Pairs 1169 and 9763 (NLI fails on unknown questions)

1169: all 20 samples name the wrong director, so the correct answer is flagged (0.999 vs 0.965).

9763: samples name four different cities, so both answers are contradicted (0.999 vs 0.990).

Task 2: Comparison with the paper

- **Paper (WikiBio, non-factual):** NLI 0.925, BERTScore 0.820, GPT-3 avg. $-\log p$ 0.832.
- **This work (HaluEval QA):** NLI 0.653 (0.910 on known questions), BERTScore 0.317, avg. $-\log p$ 0.406.
- Chance level differs: 0.73 on WikiBio, 0.50 here.
- Gain over chance: NLI $+0.20$ in the paper, $+0.15$ here ($+0.41$ on known questions). BERTScore and $-\log p$ do not reproduce.

Likely causes of the gap

- **Data:** 1,908 GPT-3 sentences vs. 400 answers; hallucinations written by ChatGPT, not by the evaluated model.
- **Compute:** GPT-3 (text-davinci-003) vs. a 3B model that cannot answer most multi-hop questions.
- **Setup:** same N, temperature and scorers, but one-word answers are compared with full-sentence samples.

Task 3: Stacked logistic regression

- 80/20 split **by question** (seed 42): 320 training and 80 test answers.
- Features: $z = (\bar{\ell}, \bar{H}, S_{\text{BERT}}, S_{\text{NLI}})$, standardised on the training set.

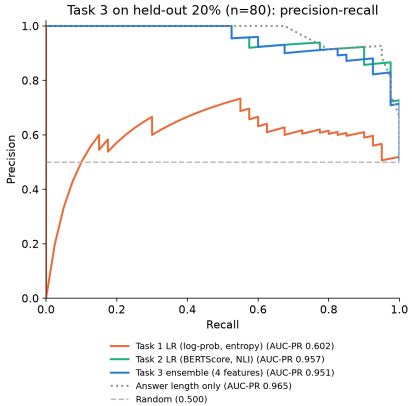
$$P(\text{hallucinated} \mid z) = \sigma(w^\top z + b) = \frac{1}{1 + e^{-(w^\top z + b)}} \qquad \text{(L2, } C = 1\text{)}$$

- Task 1, Task 2 and Task 3 evaluated on the same 80 test answers.
- 95% intervals: 1000 bootstrap resamples of the 40 test questions (seed 42).
- Repeated over 100 random splits (seeds 0–99).

Task 3: Results on the held-out 20%

Detector ($n = 80$)	AUC-PR	95% CI
Task 1: LR(log-prob, entropy)	0.602	0.53–0.72
Task 2: SelfCheck-NLI	0.651	0.56–0.75
Task 2: LR(BERTScore, NLI)	0.957	0.91–0.99
Task 3: LR on all four	**0.951**	**0.90–0.98**
Answer length only	0.965	0.94–0.99
Random	0.500	

- +0.35 over Task 1, +0.30 over NLI; 38 of 40 pairs ranked correctly.
- Level with the Task 2 regression and below length alone.
- Weights: BERTScore −3.08, NLI +0.99, log-prob −0.31, entropy −0.08. BERTScore correlates with length ($r = -0.70$).

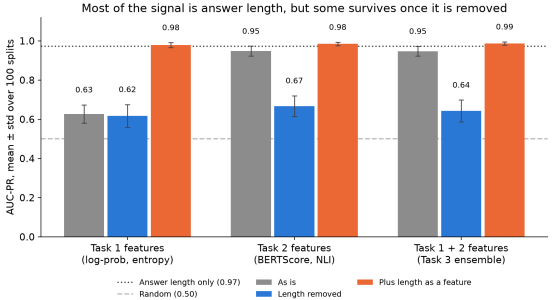


Task 3: Controlling for length

- **Length removed:** each score is replaced by its residual

$$\tilde{f} = f - (\alpha + \beta \log T)$$

- Mean over 100 splits:
  - LR(length): 0.972
  - LR(length + Task 1 + Task 2): 0.987
  - LR(Task 1 + Task 2, length removed): 0.642, 75% of pairs correct
- Mean log-probability improves most; NLI falls to chance.



Task 3: Examples

Pair 2621 (correct)

Correct: *Lewis Cass* — Task 3 score 0.06
Hallucinated: *James Buchanan was passed over for Franklin Pierce.* — Task 3 score 0.73
Lower log-probability (-5.96 vs -2.86) and higher NLI score (0.999 vs 0.77).

Pair 3657 (incorrect: labelling problem)

Correct: *American* — Task 3 score 0.74
Hallucinated: *Heart, the famous rock band, is American.* — Task 3 score 0.62
The "hallucinated" answer is in fact correct.

Pair 2060 (missed hallucination)

Correct: *Lanoxin* — Task 3 score 0.01
Hallucinated: *. . . is used in many commercial immunoassays, including several medications.* — Task 3 score 0.37
The answer makes no verifiable claim, so no sample can contradict it.

Discussion

What worked

- SelfCheck-NLI on known questions: 0.91, with no training.
- Length removed: 0.64, 75% of pairs correct.

What did not work

- Specified Task 1 scores are below chance.
- SelfCheck-BERTScore is inverted.
- Task 3 does not beat the Task 2 regression or length alone.

Why

- Hallucinations written by another model, in another style.
- A 3B model rarely knows multi-hop answers.
- Length separates the classes almost perfectly.

Limitations

- 200 questions; linear length correction.

Conclusion

- Token probabilities do not detect hallucinations on HaluEval QA.
- SelfCheckGPT is effective only on questions the model can answer.
- The Task 3 score of 0.95 is largely due to answer length; with length removed, the combined scores reach 0.64.

Code

`https://github.com/susanta75/NLP-Project-2026-Autumn`

`bash run_docker.sh` runs Task 1, 2 and 3

Thank you